

Minimum Distance classifiers operate by assigning each [42]
datapoint to the class whose class center exhibits the
minimum distance with respect to the given data point.

Correct Classification Conditions:

$$\forall \underline{x} \in X_1, \quad \|\underline{x} - \underline{\mu}_1\|_{\underline{W}}^2 < \|\underline{x} - \underline{\mu}_2\|_{\underline{W}}^2 \quad [4]$$

$$\forall \underline{x} \in X_2, \quad \|\underline{x} - \underline{\mu}_2\|_{\underline{W}}^2 < \|\underline{x} - \underline{\mu}_1\|_{\underline{W}}^2 \quad [5]$$

⊛ The correct classification conditions formulated by inequalities (4) and (5) cannot, however, directly used.

⊛ Our primary goal is to find a matrix $\underline{W}$ that transforms the feature space in such a way that the classes become as separable as possible.

⊛ Specifically, we want each class to be well-centered around its centroid and the centroids of the different classes to be far apart.

→ Achieving the previous requirement is possible by designing a cost function as a ratio of the following form :

$$J(\underline{W}) = \frac{\text{Between class Distance (J)}}{\text{Within class Distance (J)}} \quad [6]$$

→ This ratio captures two important objectives:

1: Maximizing the Between-Class Distance

We want the centroids to be as far apart as possible after applying the transformation matrix $\underline{W}$. This helps in making the two classes more distinguishable. The numerator of the cost function reflects this objective.

2: Minimizing the Within-Class Scatter

We want the points within each class to be tightly clustered around their respective centroid, making each class internally consistent and compact. The denominator of the cost function reflects this goal.

④ By maximizing the ratio, we may balance these two competing objectives, thus finding the optimal transformation matrix $\underline{W}$.

⑤ We may write that

$$J(\underline{W}) = \frac{J_{\text{between}}(\underline{W})}{J_{\text{within}}(\underline{W})} \quad [7]$$

(A): $J_{\text{between}}(\underline{W})$ measures how far apart are the centroids of the two classes, weighted by the transformation $\underline{W}$:

$$J_{\text{between}}(\underline{W}) = (\underline{\mu}_1 - \underline{\mu}_2)^T \underline{W} (\underline{\mu}_1 - \underline{\mu}_2) \quad [8]$$

CHECK EQUIVALENCE
OF QUADRATIC FORMS
AND TRACES

$$J_{\text{between}}(\underline{W}) = \text{tr} \left(\underline{W} (\underline{\mu}_1 - \underline{\mu}_2) (\underline{\mu}_1 - \underline{\mu}_2)^T \right) \quad [9]$$

Equivariance of Quadratic Forms and Traces

[#4]

$\forall \underline{v} \in \mathbb{R}^p$ and $\underline{A} \in \mathbb{R}^{p \times p}$,

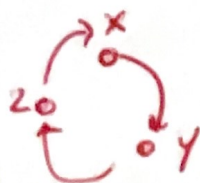
$$\underline{v}^T \underline{A} \underline{v} = \text{tr}(\underline{A} \underline{v} \cdot \underline{v}^T)$$

$$\text{tr}(\underline{A} \underline{v} \underline{v}^T) = \text{tr}(\underline{v}^T \underline{A} \underline{v}) = \underline{v}^T \underline{A} \underline{v}$$

Trace of a scalar is also a scalar

The following cyclic property of trace holds:

$$\text{tr}(\underline{X} \underline{Y} \underline{Z}) = \text{tr}(\underline{Z} \underline{X} \underline{Y}) = \text{tr}(\underline{Y} \underline{Z} \underline{X})$$



In fact, we may write that:

$$\underline{v}^T \underline{A} \underline{v} = \text{tr}(\overset{X}{\underline{v}^T} \overset{Y}{\underline{A}} \overset{Z}{\underline{v}}) = \text{tr}(\underline{v} \underline{v}^T \underline{A}) = \text{tr}(\underline{A} \underline{v} \underline{v}^T)$$

(B): $J_{\text{within}}(\underline{W})$ measures how dispersed each class is around its centroid when transformed by $\underline{W}$. It captures the total variance of points within each class relative to their centroids:

$$J_{\text{within}}(\underline{W}) = \underbrace{\sum_{\underline{x} \in X_1} (\underline{x} - \underline{\mu}_1)^T \underline{W} (\underline{x} - \underline{\mu}_1)}_{\tilde{S}_1} + \underbrace{\sum_{\underline{x} \in X_2} (\underline{x} - \underline{\mu}_2)^T \underline{W} (\underline{x} - \underline{\mu}_2)}_{\tilde{S}_2} \quad [10]$$

* Quantities $\tilde{S}_1$ and $\tilde{S}_2$ are the transformed versions of the (matrices) $\underline{S}_1$ and $\underline{S}_2$ which measure the Euclidean distance-based within-class scatter:

$$\underline{S}_1 = \sum_{\underline{x} \in X_1} (\underline{x} - \underline{\mu}_1)(\underline{x} - \underline{\mu}_1)^T \quad [11]$$

$$\underline{S}_2 = \sum_{\underline{x} \in X_2} (\underline{x} - \underline{\mu}_2)(\underline{x} - \underline{\mu}_2)^T \quad [12]$$

* Thus, we may write that:

$$\tilde{S}_1 = \sum_{\underline{x} \in X_1} (\underline{x} - \underline{\mu}_1)^T \underline{W} (\underline{x} - \underline{\mu}_1) = \sum_{\underline{x} \in X_1} \text{tr}((\underline{x} - \underline{\mu}_1)^T \underline{W} (\underline{x} - \underline{\mu}_1)) \quad [13]$$

$$\tilde{S}_2 = \sum_{\underline{x} \in X_2} (\underline{x} - \underline{\mu}_2)^T \underline{W} (\underline{x} - \underline{\mu}_2) = \sum_{\underline{x} \in X_2} \text{tr}((\underline{x} - \underline{\mu}_2)^T \underline{W} (\underline{x} - \underline{\mu}_2)) \quad [14]$$

⊛ Eqs. (13) and (14) can be re-written as:

$$\tilde{S}_1 = \sum_{\underline{x} \in X_1} \text{tr} \left(\underline{W} \cdot (\underline{x} - \underline{\mu}_1)(\underline{x} - \underline{\mu}_1)^T \right) \quad [15]$$

$$\tilde{S}_2 = \sum_{\underline{x} \in X_2} \text{tr} \left(\underline{W} \cdot (\underline{x} - \underline{\mu}_2)(\underline{x} - \underline{\mu}_2)^T \right) \quad [16]$$

It holds then: $\text{tr}(\underline{X} + \underline{Y}) = \text{tr}(\underline{X}) + \text{tr}(\underline{Y})$

⊛ Thus, Eqs. (15) and (16) can be re-expressed as:

$$\tilde{S}_1 = \text{tr} \left(\sum_{\underline{x} \in X_1} \underline{W} \cdot (\underline{x} - \underline{\mu}_1)(\underline{x} - \underline{\mu}_1)^T \right) \Rightarrow$$

$$\tilde{S}_1 = \text{tr} \left(\underline{W} \cdot \sum_{\underline{x} \in X_1} (\underline{x} - \underline{\mu}_1)(\underline{x} - \underline{\mu}_1)^T \right) = \text{tr} \left(\underline{W} \cdot \underline{S}_1 \right) \quad [17]$$

$$\tilde{S}_2 = \text{tr} \left(\sum_{\underline{x} \in X_2} \underline{W} \cdot (\underline{x} - \underline{\mu}_2)(\underline{x} - \underline{\mu}_2)^T \right) \Rightarrow$$

$$\tilde{S}_2 = \text{tr} \left(\underline{W} \cdot \sum_{\underline{x} \in X_2} (\underline{x} - \underline{\mu}_2)(\underline{x} - \underline{\mu}_2)^T \right) = \text{tr} \left(\underline{W} \cdot \underline{S}_2 \right) \quad [18]$$

⊗ We may finally write that:

$$J_{\text{within}}(\underline{\underline{W}}) = \tilde{S}_1 + \tilde{S}_2 = \text{tr}(\underline{\underline{W}} \underline{\underline{S}}_1) + \text{tr}(\underline{\underline{W}} \underline{\underline{S}}_2) \Rightarrow$$

$$J_{\text{within}} = \text{tr}(\underline{\underline{W}} \underline{\underline{S}}_1 + \underline{\underline{W}} \underline{\underline{S}}_2) = \text{tr}(\underline{\underline{W}} (\underline{\underline{S}}_1 + \underline{\underline{S}}_2)) \quad [19]$$

Letting $\underline{\underline{S}}_W = \underline{\underline{S}}_1 + \underline{\underline{S}}_2$, yields

$$J_{\text{within}}(\underline{\underline{W}}) = \text{tr}(\underline{\underline{W}} \underline{\underline{S}}_W) \quad [20]$$

⊗ According to the previous derivations, the final form of the cost function is:

$$J(\underline{\underline{W}}) = \frac{\text{tr}(\underline{\underline{W}} (\underline{\underline{\mu}}_1 - \underline{\underline{\mu}}_2)(\underline{\underline{\mu}}_1 - \underline{\underline{\mu}}_2)^T)}{\text{tr}(\underline{\underline{W}} \underline{\underline{S}}_W)} \quad [21]$$

⊗ Letting $\underline{\underline{A}} = (\underline{\underline{\mu}}_1 - \underline{\underline{\mu}}_2)(\underline{\underline{\mu}}_1 - \underline{\underline{\mu}}_2)^T$ and $\underline{\underline{B}} = \underline{\underline{S}}_W$, the distance metric learning task boils down to the following optimization problem:

$$\underline{\underline{W}}^* = \arg \max_{\underline{\underline{W}}} J(\underline{\underline{W}}) = \frac{\text{tr}(\underline{\underline{W}} \underline{\underline{A}})}{\text{tr}(\underline{\underline{W}} \underline{\underline{B}})} \quad [22]$$

SOLVE THE FOLLOWING OPTIMIZATION PROBLEM:

$$\max_{\underline{\underline{W}}} J(\underline{\underline{W}}) = \frac{\text{tr}(\underline{\underline{W}}\underline{\underline{A}})}{\text{tr}(\underline{\underline{W}}\underline{\underline{B}})} \quad (I)$$

- $\underline{\underline{W}}, \underline{\underline{A}}, \underline{\underline{B}} \in \mathbb{R}^{p \times p}$
- $\underline{\underline{A}} = (\underline{\mu}_1 - \underline{\mu}_2)(\underline{\mu}_1 - \underline{\mu}_2)^T$ is the between-class scatter matrix, representing the squared distance between class centroids.
- $\underline{\underline{B}} = \underline{\underline{S}}_W = \underline{\underline{S}}_1 + \underline{\underline{S}}_2$ is the total within-class scatter matrix, measuring the spread of each class around its centroid.
- $\underline{\underline{W}}$ is constrained to be positive semi-definite and normalized.
 $[\forall \underline{v} \in \mathbb{R}^p, \underline{v}^T \underline{\underline{W}} \underline{v} \geq 0 : \text{POSITIVE SEMI-DEFINITE MATRIX}]$
- We may impose an additional constraint on the weight-matrix $\underline{\underline{W}}$ which scales $\underline{\underline{W}}$ without affecting the optimization process

$$\max_{\underline{\underline{W}}} \text{tr}(\underline{\underline{W}}\underline{\underline{A}}) = f(\underline{\underline{W}}) \quad (\text{Objective Function})$$

$$\text{s.t.} \quad \text{tr}(\underline{\underline{W}}\underline{\underline{B}}) = 1 \quad (\text{O.P. I})$$

$$\underline{\underline{W}} \succeq 0$$

⊛ We need to define the constraint function:

$$g(\underline{\underline{W}}) = \text{tr}(\underline{\underline{W}}\underline{\underline{\beta}}) - 1 \quad (III)$$

Step: 1 Handling this constraint optimization problem within the context of Lagrangian Optimization requires forming the corresponding Lagrangian function:

$$\mathcal{L}(\underline{\underline{W}}; \lambda) = f(\underline{\underline{W}}) - \lambda \cdot g(\underline{\underline{W}}) \quad (III)$$

Eq: (III) yields:

$$\mathcal{L}(\underline{\underline{W}}; \lambda) = \text{tr}(\underline{\underline{W}}\underline{\underline{A}}) - \lambda \cdot (\text{tr}(\underline{\underline{W}}\underline{\underline{\beta}}) - 1) \quad (IV)$$

Step: 2 Apply the Karush-Kuhn-Tucker (KKT) Conditions:

(2:1) Stationarity Condition

$$\frac{\partial \mathcal{L}(\underline{\underline{W}}; \lambda)}{\partial \underline{\underline{W}}} = \underline{\underline{0}} \quad (V)$$

(2:1) Primal Feasibility Condition

$$g(\underline{\underline{W}}) = 0 \Rightarrow \text{tr}(\underline{\underline{W}}\underline{\underline{\beta}}) = 1 \quad (VI)$$

(*) The trace constraint ensures that $\underline{\underline{W}}$ is scaled appropriately.

(*) The positive semi-definite constraint is satisfied in the sense that $\underline{\underline{W}}$ has only non-negative eigenvalues.

(2:3) Dual Feasibility Condition

The dual feasibility condition requires that the Lagrangian multiplier λ is non-negative.

$$\lambda \geq 0 \quad (\text{VII})$$

(2:4) Complementarity Slackness Condition

This condition is formulated as:

$$\lambda \cdot g(\underline{W}) = 0 \quad (\text{VIII})$$

From the Stationarity Condition we have that:

$$\frac{\partial \mathcal{L}}{\partial \underline{W}} = \frac{\partial}{\partial \underline{W}} \left\{ \text{tr}(\underline{W}\underline{A}) - \lambda (\text{tr}(\underline{W}\underline{B}) - 1) \right\} = \underline{0} \Rightarrow$$

$$\frac{\partial \text{tr}(\underline{W}\underline{A})}{\partial \underline{W}} = \lambda \frac{\partial \text{tr}(\underline{W}\underline{B})}{\partial \underline{W}} \Rightarrow$$

$$\underline{A}^T = \lambda \cdot \underline{B}^T \quad (\text{Matrices } \underline{A}, \underline{B} \text{ are symmetric}) \Rightarrow$$

$$\underline{A} = \lambda^* \underline{B} \quad (IX)$$

Eq. (IX) implies that the original cost function $J(\underline{W})$ assumes its maximum value for λ^* ,

$$J(\underline{W}^*) = \frac{\text{tr}(\underline{W}^* \underline{A})}{\text{tr}(\underline{W}^* \underline{B})} = \frac{\text{tr}(\underline{W}^* \lambda^* \underline{B})}{\text{tr}(\underline{W}^* \underline{B})} = \frac{\lambda^* \text{tr}(\underline{W}^* \underline{B})}{\text{tr}(\underline{W}^* \underline{B})} = \lambda^* \quad (X)$$

⑥ Eq. (IX) defines a generalized eigenvalue equation of [11]
the form:

$$\underline{A} \cdot \underline{V} = \lambda^* \underline{B} \underline{V} \quad (x1)$$

⑦ In fact, when $\underline{B}$ is invertible, Eq. (x1) can be transformed to an ordinary eigenvalue equation as:

$$\underline{B}^{-1} \underline{A} \cdot \underline{V} = \lambda^* \underline{V} \quad (x11)$$

⑧ Thus, according to Eq. (X), the cost function $J(\underline{W})$ may acquire its maximum value by selecting λ^* as the maximum eigenvalue of matrix $\underline{M} = \underline{B}^{-1} \underline{A}$ and

$$\underline{M} \cdot \underline{V}_{\max} = \lambda_{\max} \cdot \underline{V}_{\max} \quad (x111)$$

⑨ Focusing on the Primal Feasibility Condition given by Eq. (VI):

$$\text{tr}(\underline{W} \underline{B}) = 1$$

⑩ Apparently, there are infinitely many matrices $\underline{W}$ that satisfy Eq. (vi). We may incorporate additional constraints on the matrix $\underline{W}$ so that it reflects the eigenstructure of $\underline{B}$ as:

$$\underline{W} = c \underline{B}^{-1} \quad (x1v)$$

⑪ Under these conditions, Eq. (vi) yields: $\text{tr}(c \underline{B}^{-1} \underline{B}) = 1 \Rightarrow$

$$\text{tr}(c \underline{I}) = 1 \Rightarrow c \cdot e = 1 \Rightarrow c = e^{-1} \quad (x1v)$$

⑫ Finally, we get: $\underline{W}^* = \frac{1}{e} \underline{B}^{-1} \Rightarrow \underline{W}^* = \frac{1}{e} \underline{S}_w^{-1} \Rightarrow$

$$\underline{W}^* = \frac{1}{e} (\underline{S}_1 + \underline{S}_2)^{-1} \quad (x1vi)$$